

The Variance Risk Premium: Model-Free Measurement, Term Structure, and the Economics of Selling Volatility

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Abstract

Option-implied variance on the S&P 500 tends to run higher than the variance that ends up realized. In this paper, we derive the model-free measurement apparatus underlying that fact from first principles. The derivation starts with log-contract replication of realized variance via Itô's lemma. We then prove the Carr-Madan spanning identity. Finally, we build the exact discretization bridge to the CBOE's VIX formula. We then measure the resulting variance risk premium over 22 years of daily S&P 500 and VIX data (2004–2026). To do this, we use a forward-window realized-variance convention argued to be the only construction consistent with the premium's economic meaning. The premium averages approximately 3.5 volatility points. It is positive on 84% of sample days, and has a left-skewed distribution (skewness -7 in variance units). Additionally, its sign inverts during the global financial crisis, though not in every crisis regime in the sample. A VIX futures term structure built over 2007–2026 shows contango on 82% of days. It mirrors the same premium from a second instrument. A short-volatility strategy constructed from that curve illustrates the premium's economic payoff profile which is steady accumulated gains punctuated by catastrophic single-day losses. Section 6 relates these findings to a companion paper on rough volatility. It draws the line between what path roughness explains about this premium and what it leaves untouched.

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1 Introduction

Option-implied variance on the S&P 500 exceeds the variance that is later realized. Over the 2004–2026 sample that we study in this paper, the gap averages approximately 3.5 volatility points and is positive on 84% of trading days. On the majority of days, the market’s option-implied forecast of the next month’s variance is higher than what actually occurs. This gap is the difference between risk-neutral and physical expectations of the same quadratic variation. It is the **variance risk premium**, and it is the object we measure in this paper.

This gap should, in principle, get competed away by arbitrage. It doesn’t, and the usual explanation is an insurance argument. Realized variance spikes almost exclusively in market crashes. Those are exactly the states where a representative investor’s marginal utility of wealth is highest. A claim that pays off in those states is worth more to investors than its actuarial value. In fact, the payoff arrives when it is needed most. Investors buy variance exposure through options, variance swaps, or the log-contract replication we derive in this paper. In effect they are buying crash insurance, and they’re willing to pay above fair value for it. On the other hand, investors who sell that exposure earn a steady premium in exchange for carrying an infrequent left tail. Two features of the data back this up. The distribution of the premium is sharply left-skewed (skewness of -7 in variance units, driven by a small number of days where realized variance far exceeds what was priced). And the premium’s *sign* flips during the global financial crisis, the one regime in the sample where insurance actually gets called on.

This paper makes three contributions: a rigorous way to measure implied variance, a careful construction of realized variance to set against it, and an account of what mechanism could plausibly generate a persistent wedge between the two. First, we derive the model-free measurement apparatus from first principles instead of citing it secondhand. The derivation starts with log-contract replication of realized variance. It builds a complete proof of the Carr-Madan spanning identity. It then arrives at the model-free implied variance formula and reconciles it exactly with the CBOE’s published VIX methodology. Every discretization choice in the VIX, strike spacing, the K_0 correction, truncation, discounting, is traced back to the continuous-time theory it approximates. Second, we construct the variance risk premium across 22 years of daily data, using a forward-window convention we argue is the only construction consistent with what a risk premium actually means. Alongside it, we build the VIX futures term structure (2007–2026) and relate it to the same premium through its contango behavior. We also backtest a simple short-volatility strategy to make the premium’s economic content concrete. Third, Section 6 connects these findings to the rough volatility literature through a companion paper. It separates what path roughness explains about these results from what it leaves unexplained.

The headline results, for a reader who takes away only the numbers: the mean variance risk premium is approximately 3.5 volatility points, positive on 84% of sample days, with a skewness of -7 in variance units driven by a small number of extreme realized-variance outcomes. The VIX futures term structure is in contango 82% of days — the futures-market mirror of the same premium, and a close match to the 84% positive-VRP frequency measured directly. The premium is regime-dependent in exactly the way the insurance argument predicts: it turns negative during the global financial crisis, when crash insurance is called upon to pay out. The Covid regime, discussed in Section 5.3, follows a different pattern and is treated separately rather than folded into the crisis-inversion story here. A short-volatility strategy constructed directly from the VIX futures curve grinds upward by approximately 400 volatility points over 19 years, punctuated by two severe single-day losses (-17.6 vol points on February 5, 2018, and -19.2 vol points on March 16, 2020) that account for a disproportionate share of the strategy’s total risk and illustrate directly the left-tail outcome this paper’s insurance framing predicts.

This paper doesn't propose new theory: the derivations in Sections 2 and 3 are known results in the literature. This paper works through them completely and rigorously instead of citing and skipping past them. We do that so that every step from Itô's lemma to the published VIX formula is visible and checkable. The empirical contribution is a reproducible measurement over an unusually long and crisis-rich sample. It offers no new pricing theory and no new statistical test. Three scope limitations matter enough to state up front. The theoretical derivation assumes a continuous price process with no jumps. Section 2.1 spells out what that costs instead of leaving it implicit. The realized-variance estimator uses daily data, trading some statistical efficiency for robustness and data availability over a 22-year sample (Section 3.3). Finally, the short-volatility backtest in Section 5 illustrates the premium's economic sign and magnitude only. It excludes transaction costs, margin requirements, and financing costs, so it isn't a claim about a tradeable, net-of-cost strategy.

This paper draws on two separate literatures and doesn't claim to originate either one. The model-free replication theory follows Carr and Madan (1998), Britten-Jones and Neuberger (2000), and Demeterfi et al. (1999). The measurement and economic interpretation of the variance risk premium follows Carr and Wu (2009) and Bakshi and Kapadia (2003). The rough volatility connection in Section 6 follows Gatheral et al. (2018) and builds directly on this author's own prior work (Amargos, 2026). What this paper adds is a complete derivation of the replication theory, from Itô's lemma all the way to the published VIX formula. It also provides an updated 22-year measurement of the premium that spans two major crisis regimes the original literature could only partially observe.

The remainder of the paper is organized as follows. In Section 2 we derive the model-free measurement apparatus, from the log-contract identity to the exact bridge to the CBOE VIX formula. In Section 3 we construct the realized-variance estimator and argue for the forward-window convention. Section 4 defines the variance risk premium and its futures-market counterpart, the term-structure slope. Section 5 measures both empirically over 2004–2026, including the term structure, regime comparisons, and the short-volatility backtest. In Section 6 we connect these findings to the rough volatility literature via the companion paper. Section 7 concludes.

2 Mathematical Foundations

2.1 Setup and Assumptions

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{Q})$ under a risk-neutral measure $\mathbb{Q}$. The underlying asset price S_t is assumed to be a strictly positive and continuous semimartingale, following

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t,$$

where W_t is a $\mathbb{Q}$ -Brownian motion and μ_t, σ_t are adapted processes; σ_t in particular may itself be stochastic. Three assumptions are made explicit, since each one is doing real work in what follows:

1. **Continuous paths, no jumps.** S_t has no discontinuities, since rough volatility models are themselves continuous but jump-diffusion models for the underlying are not.
2. **Frictionless markets with a continuum of strikes.** For every maturity T , European options trade at every strike $K \in (0, \infty)$, with no bid-ask spread, transaction costs, or liquidity constraints. This is the idealization that the CBOE methodology approximates with a discrete strike grid. The gap between the two is the subject of Section 2.5.

3. **Deterministic (or zero) risk-free rate.** For expositional clarity the derivation in 2.2–2.4 is carried out with $r = 0$, so that forward price equals spot price and option prices coincide with discounted expected payoffs. The deterministic-rate case is recovered by reintroducing discount factors e^{-rT} throughout, which is the role played by the e^{RT} term in the CBOE formula (Section 2.5).
4. **No-arbitrage drift restriction under $\mathbb{Q}$.** Since $\mathbb{Q}$ is a risk-neutral measure, the discounted price process must be a $\mathbb{Q}$ -martingale, which forces $\mu_t = r = 0$ in the dynamics above. This is what makes S_t itself a $\mathbb{Q}$ -martingale in Section 2.4, and is what gives the dynamic-hedge term zero expectation.

A note on jumps. If S_t is allowed to jump, the identity linking option prices to expected quadratic variation breaks down. The model-free variance swap rate then measures expected quadratic variation *plus* a correction term. That correction is driven by the third moment of the jump distribution. The reason is that the Itô/Taylor expansion used below no longer captures the full local behavior of $\ln S_t$ at a jump. Carr and Wu (2009) derive this correction explicitly and show it can be economically significant in the presence of a pronounced jump-risk premium. This paper works entirely in the continuous case. The jump correction is left as an acknowledged limitation.

2.2 The Log Contract and Realized Variance

The starting point of the entire model-free approach is a single application of Itô’s lemma to $f(x) = \ln x$. Since $f'(x) = 1/x$ and $f''(x) = -1/x^2$, Itô’s formula gives

$$d \ln S_t = \frac{dS_t}{S_t} - \frac{1}{2} \sigma_t^2 dt.$$

Integrating from 0 to T and rearranging:

$$\int_0^T \sigma_t^2 dt = 2 \left(\int_0^T \frac{dS_t}{S_t} - \ln \left(\frac{S_T}{S_0} \right) \right). \quad (1)$$

Every term in (1) has a trading interpretation, which is the point of the identity.

- $\int_0^T \frac{dS_t}{S_t}$ is the terminal P&L of a **dynamic delta-hedging strategy**: hold $1/S_t$ shares of the underlying at every instant t , continuously rebalanced, financed with borrowing/lending at the (zero) risk-free rate. Because it requires continuous rebalancing to $1/S_t$ shares, it is a self-financing strategy rather than a static one.
- $-\ln(S_T/S_0)$ is the terminal payoff of a **static short position in a “log contract”**, a hypothetical instrument paying $\ln(S_T/S_0)$ at maturity, which the investor is short.

So (1) says: **realized variance is the payoff of a replicable trading strategy**, long the dynamic delta-hedge and short the static log contract. That payoff is pinned down by a hedging argument. A variance swap can therefore be priced model-independently. Variance itself becomes a *tradeable* asset rather than a number you can only compute after the fact.

The factor of 2. The coefficient of 2 in (1) originates directly from the second-order term in Itô’s lemma, specifically from $f''(x) = -1/x^2$ carrying a $\frac{1}{2}$ prefactor in the Taylor expansion, which the rearrangement above inverts and doubles. That factor is forced by the second derivative of the logarithm. The same factor of 2 reappears unchanged in the model-free implied variance formula

of Section 2.4, and from there directly in the CBOE's $2/T$ prefactor in the VIX formula (Section 2.5). When we trace this constant from here to the VIX white paper we confirm the derivation has not silently picked up an extra factor anywhere along the way.

The remaining task is to show that the static leg, meaning the log contract, can itself be replicated by a portfolio of listed options, since no such contract trades in practice. That is the job of the Carr–Madan spanning result.

2.3 The Carr–Madan Spanning Theorem

Claim 1. *For any twice continuously differentiable function f and any constant $\kappa > 0$,*

$$f(S_T) = f(\kappa) + f'(\kappa)(S_T - \kappa) + \int_0^\kappa f''(K)(K - S_T)^+ dK + \int_\kappa^\infty f''(K)(S_T - K)^+ dK. \quad (2)$$

This says that any smooth payoff can be spanned as follows: a bond paying $f(\kappa)$, a forward position of size $f'(\kappa)$ struck at κ , a continuum of out-of-the-money puts (strikes below κ) weighted by $f''(K) dK$, and a continuum of out-of-the-money calls (strikes above κ) weighted by $f''(K) dK$.

Proof. Fix a realization S_T and split into two cases.

Case $S_T \geq \kappa$. By the fundamental theorem of calculus,

$$f(S_T) - f(\kappa) = \int_\kappa^{S_T} f'(u) du.$$

Subtract and add the linear term:

$$f(S_T) - f(\kappa) - f'(\kappa)(S_T - \kappa) = \int_\kappa^{S_T} [f'(u) - f'(\kappa)] du.$$

Write $f'(u) - f'(\kappa) = \int_\kappa^u f''(K) dK$ and substitute:

$$= \int_\kappa^{S_T} \int_\kappa^u f''(K) dK du.$$

Swap the order of integration. The region is $\{(K, u) : \kappa \leq K \leq u \leq S_T\}$, so integrating K first over $[\kappa, S_T]$ and, for each fixed K , u over $[K, S_T]$:

$$= \int_\kappa^{S_T} f''(K) \left(\int_K^{S_T} du \right) dK = \int_\kappa^{S_T} f''(K)(S_T - K) dK.$$

Since $(S_T - K)^+ = 0$ for $K > S_T$, this integral is unchanged if extended to $\int_\kappa^\infty f''(K)(S_T - K)^+ dK$. This recovers exactly the call-side term of (2), with the put-side term contributing zero (its integrand $(K - S_T)^+$ vanishes for all $K \leq \kappa \leq S_T$).

Case $S_T \leq \kappa$. By an identical argument applied to $f(\kappa) - f(S_T) = \int_{S_T}^\kappa f'(u) du$,

$$f(S_T) - f(\kappa) - f'(\kappa)(S_T - \kappa) = \int_{S_T}^\kappa f''(K)(K - S_T) dK = \int_0^\kappa f''(K)(K - S_T)^+ dK,$$

with the call-side term contributing zero for the symmetric reason.

The two cases partition all possible outcomes and each recovers one side of (2) while annihilating the other. Hence, the identity holds unconditionally, for every realization of S_T . Taking f to be C^2 was used only to guarantee that f'' exists as an ordinary function under the integral sign. $\square$

Interpretation. Equation (2) is a static replication result. Any European-style payoff $f(S_T)$, however nonlinear, can be manufactured today by trading a bond, a forward, and a continuum of vanilla options at every strike, with no rebalancing required before expiry. All the “dynamics” implicit in a nonlinear payoff are absorbed into the second-derivative weighting $f''(K)$ applied to the option strip. This is the tool needed to replicate the log contract from Section 2.2, since $f(x) = \ln x$ (or $-\ln x$) is a smooth, strictly concave function to which (2) applies directly.

2.4 Model-Free Implied Variance

Apply the spanning identity to $f(x) = -\ln x$, the payoff of the static short log-contract position identified in Section 2.2. Here $f'(x) = -1/x$ and $f''(x) = 1/x^2$, so (2) gives

$$-\ln S_T = -\ln \kappa - \frac{1}{\kappa}(S_T - \kappa) + \int_0^\kappa \frac{1}{K^2}(K - S_T)^+ dK + \int_\kappa^\infty \frac{1}{K^2}(S_T - K)^+ dK.$$

Since $f''(K) = 1/K^2 > 0$ for $f = -\ln x$, replicating the short log contract requires a strictly long position in options at every strike, weighted by $1/K^2$. This means buying convexity, not selling it. That matches the economic intuition from Section 2.2: the investor short the log contract is short a concave payoff, and hedging a short concave position requires being long convex instruments.

Choosing $\kappa = F$. Set κ equal to the forward price F for maturity T . Under $\mathbb{Q}$ (and with $r = 0$, so $F = \mathbb{E}^\mathbb{Q}[S_T] = S_0$), the linear term has zero expectation:

$$\mathbb{E}^\mathbb{Q}\left[-\frac{1}{F}(S_T - F)\right] = -\frac{1}{F}(\mathbb{E}^\mathbb{Q}[S_T] - F) = 0.$$

Centering at the forward is the unique choice that kills the forward-exposure term, leaving only observable option prices. Taking $\mathbb{Q}$ -expectations of the identity above:

$$\mathbb{E}^\mathbb{Q}[-\ln S_T] = -\ln F + \int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK, \quad (3)$$

where $P(K) = \mathbb{E}^\mathbb{Q}[(K - S_T)^+]$ and $C(K) = \mathbb{E}^\mathbb{Q}[(S_T - K)^+]$ are (undiscounted, since $r = 0$) expected option payoffs.

Combining with Section 2.2. Take $\mathbb{Q}$ -expectations of (1):

$$\mathbb{E}^\mathbb{Q}\left[\int_0^T \sigma_t^2 dt\right] = 2 \mathbb{E}^\mathbb{Q}\left[\int_0^T \frac{dS_t}{S_t}\right] - 2 \mathbb{E}^\mathbb{Q}[\ln(S_T/S_0)].$$

Under $\mathbb{Q}$ with zero rates, S_t is itself a $\mathbb{Q}$ -martingale, so the dynamic-hedge term has zero expectation: $\mathbb{E}^\mathbb{Q}\left[\int_0^T dS_t/S_t\right] = 0$. Using (3) and $F = S_0$:

$$\mathbb{E}^\mathbb{Q}\left[\int_0^T \sigma_t^2 dt\right] = 2 \left(\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right). \quad (4)$$

The central result of the section is this: **expected realized variance over $[0, T]$ equals the price of a specific, model-independent portfolio of listed options**, and no assumption on the dynamics of σ_t was used anywhere in the derivation. Annualizing gives the **model-free implied variance**:

$$\sigma_{\text{MFIV}}^2 = \frac{2}{T} \left(\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right). \quad (5)$$

Two things are worth noting before moving to the VIX itself. First, with a nonzero deterministic rate r , option *prices* are discounted expected payoffs, $P(K) = e^{-rT} \mathbb{E}^\mathbb{Q}[(K - S_T)^+]$; recovering the

undiscounted expectation used above requires multiplying market prices by e^{rT} , which is the e^{rT} factor that appears in the CBOE formula in Section 2.5. Second, (5) requires a continuum of strikes, which does not exist in any real market; the next subsection is devoted to bridging that gap.

2.5 From Theory to the VIX

CBOE (2019) computes, for each of the two nearest SPX option maturities straddling 30 days,

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2,$$

with $VIX = 100 \times \sqrt{\sigma^2}$ (interpolated between the two maturities to a constant 30-day horizon). Every deviation from (5) in this formula corresponds to a specific real-world constraint the theory above assumed away:

(i) Discrete strikes. The integral $\int f''(K) \cdot (\text{price}) dK$ becomes a sum over listed strikes, with $\Delta K_i = (K_{i+1} - K_{i-1})/2$ acting as the trapezoidal-rule weight for strike K_i . This numerical quadrature approximation to (5) has accuracy that depends entirely on how densely strikes are listed near the money. A coarser grid introduces discretization error directly into the variance estimate.

(ii) Centering at K_0 , not F . The theory in Section 2.4 requires centering exactly at the forward price F , which almost never coincides with a listed strike. CBOE instead centers the sum at K_0 , the strike immediately below F , and both a put and a call quote at K_0 are averaged into the sum. The term $-\frac{1}{T}(F/K_0 - 1)^2$ is the second-order Taylor remainder correcting for the resulting mismatch between K_0 and F . It is small whenever K_0 is close to F (fine strike grid near the money), and zero if $K_0 = F$.

(iii) Strike truncation. In practice, as strikes move far out of the money, bids fall to zero and no further strikes are quoted. Truncating the integral at the last valid strike drops a small amount of tail probability mass. It biases the variance estimate slightly downward. This bias is worst in calm, low-volatility markets, because that is when far-OTM options are least liquid and get truncated earliest. It is an important caveat for anyone using the VIX to study low-volatility regimes.

(iv) The e^{rT} factor. As noted at the end of Section 2.4, this converts discounted market midquotes $Q(K_i)$ into the undiscounted expected payoffs that the theoretical derivation requires, restoring consistency between the discrete formula and (5).

Closing the chain. These four adjustments only affect measurement precision. They don't change what's being measured. $VIX_t^2/100^2$ is the CBOE's discretized estimate of σ_{MFIV}^2 , the risk-neutral expectation of annualized quadratic variation over the next 30 days, computed without reference to any option pricing model. Section 4 will need this quantity on the implied-variance side of the variance risk premium, set against a realized-variance estimator that we construct in Section 3.

3 Realized Variance

3.1 Quadratic Variation and the Convergence Result

Section 2 showed that expected variance under $\mathbb{Q}$ can be recovered model-independently from a strip of option prices. The other side of the variance risk premium requires a physical-measure estimate of how much variance actually occurred. This section builds that estimator.

For a partition $0 = t_0 < t_1 < \dots < t_n = T$ of $[0, T]$, define realized variance over the partition as the sum of squared log returns:

$$RV_{[0,T]} = \sum_{i=0}^{n-1} (\ln S_{t_{i+1}} - \ln S_{t_i})^2.$$

Theorem 1 (Convergence to quadratic variation). *If $\ln S_t$ is a continuous semimartingale, then as the mesh of the partition $\max_i(t_{i+1} - t_i) \rightarrow 0$,*

$$\sum_{i=0}^{n-1} (\ln S_{t_{i+1}} - \ln S_{t_i})^2 \xrightarrow{\mathbb{P}} \langle \ln S \rangle_T = \int_0^T \sigma_t^2 dt.$$

The result is standard in stochastic calculus. See Protter (2005) or Jacod and Shiryaev (2003) for the general semimartingale convergence theory.

Intuition. Write a single increment using the semimartingale decomposition, $\ln S_{t_{i+1}} - \ln S_{t_i} \approx \mu_{t_i} \Delta t_i + \sigma_{t_i} \Delta W_i$, where $\Delta t_i = t_{i+1} - t_i$ and $\Delta W_i \sim \mathcal{N}(0, \Delta t_i)$ is the Brownian increment. Squaring:

$$(\ln S_{t_{i+1}} - \ln S_{t_i})^2 \approx \mu_{t_i}^2 \Delta t_i^2 + 2\mu_{t_i} \sigma_{t_i} \Delta t_i \Delta W_i + \sigma_{t_i}^2 (\Delta W_i)^2.$$

The first term is $O(\Delta t_i^2)$. The cross term has mean zero and variance $O(\Delta t_i^3)$, so it washes out in the sum. The last term is the one that survives. $(\Delta W_i)^2$ has mean Δt_i and fluctuates around it with standard deviation $O(\Delta t_i)$, so summing $\sigma_{t_i}^2 (\Delta W_i)^2$ over a fine partition concentrates around $\sum \sigma_{t_i}^2 \Delta t_i \rightarrow \int_0^T \sigma_t^2 dt$ by a law-of-large-numbers argument across increments. Squaring the increments is what does the work here. In fact, it is the operation that kills the linear-in- Δt drift and cross terms relative to the diffusion term, leaving only accumulated variance in the limit.

Connecting Sections 2 and 3. Section 2 established that the option strip prices $\mathbb{E}^{\mathbb{Q}}[\int_0^T \sigma_t^2 dt]$, which is expected quadratic variation under the risk-neutral measure, looking forward. This section constructs an estimator of the same object $\int_0^T \sigma_t^2 dt$, but realized under the physical measure $\mathbb{P}$, looking backward from the vantage point of T . The variance risk premium is the difference between these two views of the same quantity: what the market paid to insure against quadratic variation it had not yet seen, versus what quadratic variation actually arrived. Every empirical result in Section 5 measures the gap between these two versions of the same random variable.

3.2 Why the Drift Doesn't Matter

The intuition in 3.1 can be sharpened into an order-of-magnitude statement that has a direct practical consequence: it tells us not to bother demeaning returns before computing realized variance.

For a single increment over Δt , the drift contributes a term of size

$$(\mu \Delta t)^2 = O(\Delta t^2),$$

while the diffusion contributes

$$\mathbb{E}[\sigma^2 (\Delta W)^2] = \sigma^2 \Delta t = O(\Delta t).$$

At daily frequency, $\Delta t \approx 1/252 \approx 0.004$, so $\Delta t^2 \approx 1.6 \times 10^{-5}$, smaller than Δt itself by another factor of Δt . The drift's contribution to squared daily returns is second-order small relative to the diffusion's, and this gap only widens as Δt shrinks further under intraday sampling. Section 3.1 made this claim informally. Here it's the actual order of magnitude, confirming that drift washes out of realized variance in the continuous-time limit and is already negligible at daily frequency.

Practical corollary: don’t demean. A tempting “improvement” is to subtract the sample mean return before squaring, $RV = \sum_i (r_i - \bar{r})^2$, on the logic that this isolates the variance from the drift more cleanly. This is a mistake at daily frequency over short windows. The sample mean $\bar{r}$ is estimated with substantial noise from only 21 observations, and subtracting a noisy estimate of an already-negligible drift term *adds* estimation variance to the realized-variance estimator for no meaningful reduction in bias. The standard practice in the realized-volatility literature (e.g. Andersen et al. (2003)) is exactly this: use raw squared returns, uncentered, at daily or higher frequency. This paper follows that convention.

3.3 The Estimator

The realized variance estimator used throughout this paper is

$$RV_t = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2,$$

where $r_{t+i} = \ln(S_{t+i}/S_{t+i-1})$ is the daily close-to-close log return of the S&P 500 index. Each design choice is defended in turn.

Close-to-close, daily, not intraday. Higher-frequency (e.g. 5-minute) realized variance is known to be a more statistically efficient estimator of quadratic variation. It converges faster as sampling frequency increases. This is well established in the realized-volatility literature (Andersen et al., 2003). It is not used here for two reasons. The first is that intraday microstructure noise biases naive high-frequency realized variance upward once sampling gets too fine. Fixing that requires noise-correction techniques, and that’s a layer of methodology this paper doesn’t need. The second is that daily close data for the full 2004–2026 sample is far easier to get and more reliable than clean intraday data over the same span. Daily close-to-close realized variance ends up noisier, but it’s unbiased, it’s the standard choice in the literature, and it’s good enough for measuring a monthly-horizon premium.

21 trading days. This window is chosen to match the VIX’s 30-calendar-day horizon. 30 calendar days contains approximately 21 trading days once weekends (and, to a first approximation, holidays) are excluded. This alignment is what makes RV_t over $[t, t + 21]$ the right object to net against VIX_t^2 . Mismatched horizons would contaminate the VRP with a pure term-structure effect having nothing to do with a risk premium.

Annualization by $\times 252$. Multiplying the sum of 21 daily squared returns by 252/21 converts a 21-trading-day variance into annualized units, matching the annualized convention of $(VIX_t/100)^2$. 252 is the standard convention for trading days per year used throughout the volatility literature.

No demeaning. Per the argument in Section 3.2, raw squared returns are used with no subtraction of the sample mean.

Units. RV_t as defined is an annualized variance (units of return² per year). Its square root, $\sqrt{RV_t}$, is annualized volatility in the same units as $VIX_t/100$. Section 4 is explicit about never mixing variance and volatility units when defining the VRP, a discipline enforced throughout.

3.4 The Forward-Looking Convention

Definition. The realized variance comparable to VIX_t is the *forward* realized variance over $[t, t+21]$, the 21 trading days following date t . It is not the trailing realized variance over $[t - 21, t]$.

It is the single most important modeling choice in the paper, and deriving it from first principles is worth the space rather than simply asserting it.

The argument. Per Section 2, VIX_t is built entirely from option prices observed at date t , for options expiring at $t + 30$ calendar days. It is a statement about the market’s risk-neutral expectation of quadratic variation over the future window $[t, t + T]$.

For the VRP to mean anything economically, it has to be a premium paid for insurance against an outcome that is still uncertain. So the realized-variance leg of the comparison has to reference that same future window.

Suppose instead VIX_t were compared against trailing realized variance over $[t - 21, t]$. That would compare a forward-looking price against a quantity already fully known at date t . There is no uncertainty left to price in the trailing window. A large trailing-RV-based “premium” would then just measure how much realized volatility had recently persisted, volatility clustering. That is a statement about how σ_t moves. It has nothing to do with compensation for bearing future variance risk.

Insurance priced against something still uncertain is a different thing from insurance compared against a claim already settled. Only the first one is a premium in any meaningful economic sense.

There’s a practical symptom of this too. Trailing RV is itself a smoothed, backward-looking object, so a trailing-RV comparison would look mechanically smoother and more persistently positive than the true VRP. That smoothness would come from comparing to the wrong window. It wouldn’t reflect anything about a real risk premium.

Consequences of the forward convention.

- *Endpoint caveat.* RV_t needs 21 trading days after t to be computed. So it can’t be computed for the last 21 trading days of the sample, those windows run past the last price observation we have. This means the VRP series ends 21 trading days before the raw data actually ends. Section 5’s data appendix notes this gap.
- *Overlapping observations and autocorrelation.* RV_t and RV_{t+1} share 20 of their 21 underlying daily returns. So consecutive daily VRP observations overlap a lot, and they end up strongly autocorrelated just from how the estimator is built. That is a mechanical fact about the construction. It doesn’t mean the premium itself actually persists day to day. This matters a lot in Section 5. If we computed summary statistics or standard errors treating these overlapping observations as independent, we’d seriously understate how uncertain the estimates really are.
- *Lead-lag as an internal consistency check.* Forward realized variance tends to spike *before* VIX_t does, around big market drawdowns, which is a useful way to check the construction was done right. That makes sense, since RV_t for a date just before a crash already looks forward into the crash window, while VIX_t on that same date only reflects what the market expected ex ante, not what actually happened. We can see this lead-lag pattern directly in the constructed series (Figure 1, Section 5). It’s not really a finding on its own but more a check. It confirms the forward-window convention was implemented correctly, since a trailing-RV construction wouldn’t show this pattern at all.

4 The Variance Risk Premium

4.1 Definition

With an implied-variance measure from Section 2 and a realized-variance estimator from Section 3 in hand, the variance risk premium is defined as

$$VRP_t \equiv \left(\frac{VIX_t}{100} \right)^2 - RV_{t \rightarrow t+21}. \quad (6)$$

Both terms are in annualized variance units and it matters here more than anywhere else because the two most common mistakes in this literature are (i) mixing variance and vol-point units in the same expression, and (ii) treating the vol-point version of the VRP as if it were simply the square root of the variance-point version.

VRP_t as defined in (6) is the **ex post, realized-payoff** definition of the premium: what an investor who sold a 30-day variance swap at the fair strike VIX_t^2 would earn, per unit of variance notional, once the 21-day window resolves. A positive realization is a cash payoff the seller actually collects. It comes from an actual trade and not a model.

The vol-points version, and its warning. For readability, results are also reported in vol points as $VIX_t - 100\sqrt{RV_{t \rightarrow t+21}}$. This is a difference of volatilities, not variances, and both legs sit on the same 0–100 index scale. The two only coincide to a first-order approximation, and only when the variance gap is small relative to the level of variance itself. The alternative expression, $100\sqrt{(VIX_t/100)^2 - RV_{t \rightarrow t+21}}$, is undefined on any day realized variance exceeds implied variance, since that would require taking the square root of a negative number. This is a dangerous kind of bug. A script computing $\sqrt{IV^2 - RV}$ will silently produce NaN whenever realized vol exceeds implied vol, with no visible error, and a careless implementation could drop those days from a summary statistic without anyone noticing. Every vol-points figure in this paper is computed as the simple difference $VIX_t - 100\sqrt{RV_{t \rightarrow t+21}}$ instead, which is always well-defined.

Sign convention. $VRP_t > 0$ means implied variance exceeded later realized variance. The seller of variance insurance collected more in premium than they ultimately paid out in that realized window. $VRP_t < 0$ means the reverse. Realized variance came in above what was priced, and the insurance seller took a loss on that window.

Endpoint caveat. As established in Section 3.4, $RV_{t \rightarrow t+21}$ requires 21 trading days of data after t , so VRP_t is unmeasurable for the final 21 trading days of any sample. The same caveat applies to every VRP figure and table in Section 5; it isn't repeated at each one.

The expectation interpretation. Taking physical-measure expectations of (6),

$$\mathbb{E}^{\mathbb{P}}[VRP_t] > 0 \iff \mathbb{E}^{\mathbb{Q}}[QV] > \mathbb{E}^{\mathbb{P}}[QV],$$

where QV denotes quadratic variation over the matching window. The premium is the wedge between two measures applied to the same random variable. VIX_t^2 is quadratic variation priced under $\mathbb{Q}$, $RV_{t \rightarrow t+21}$ is an estimate of the same quadratic variation realized under $\mathbb{P}$, and a systematic positive gap between the two is what a risk premium means in asset pricing.

Why doesn't this wedge get competed down to zero? The standard answer, developed by Carr and Wu (2009) and backed up by the broader variance-premium evidence in Bakshi and Kapadia (2003), comes down to the state-dependence of marginal utility: realized variance spikes are concentrated almost entirely in market crashes and periods of acute stress. Those are exactly the states where a representative investor's marginal utility of wealth is highest. A payoff that is large in high-marginal-utility states commands a price above its actuarial (physical-probability-weighted) value. It is because investors are willing to pay a premium, in expectation-losing terms,

for a claim that pays off when they need the money most. Being long variance is exactly this kind of payoff. Being short variance is selling that insurance and collecting the premium in exchange for bearing the tail risk. The wedge in (6) is the standard risk-premium story. It is applied to the second moment of returns instead of the first. It reflects compensation for risk, not mispricing.

4.2 Measurement Conventions and Pitfalls

The forward-window convention established in Section 3.4 governs (6) directly: $RV_{t \rightarrow t+21}$ must reference the window following t , matching the horizon priced into VIX_t . Get that wrong and the resulting series measures volatility persistence, not a risk premium.

Overlapping observations. $RV_{t \rightarrow t+21}$ and $RV_{t+1 \rightarrow t+22}$ share 20 of 21 underlying daily returns, so the daily VRP_t series inherits strong, mechanical MA(20)-type autocorrelation just from how it’s built. This matters for inference: naive t -statistics computed on daily VRP observations, treating them as independent, will seriously overstate statistical significance, since the effective number of independent observations is far smaller than the daily sample count suggests. The standard fixes are Newey-West (HAC) standard errors with a lag length tied to the 21-day overlap, or subsampling to non-overlapping 21-day blocks at the cost of a much smaller sample. This paper’s empirical work in Section 5 leans on descriptive statistics, visual evidence, and regime comparisons rather than formal hypothesis tests. An inference procedure built for independent observations is the wrong tool for overlapping data, and it’s worth saying that plainly for any reader tempted to compute their own t -statistic on daily VRP.

Units discipline. Every table and figure in Section 5 reports results in both annualized variance units and vol points, and the two are never combined into a single number. A mean VRP of “0.0077” (variance units) and “3.5 vol points” are two readings of the same result, and this paper says so plainly wherever both appear together, since that’s exactly the confusion the warning in 4.1 was meant to head off.

This correspondence holds for the pooled sample, but not always within a single regime. Table 2 shows a variance-unit mean of -0.011 for the GFC alongside a vol-point mean of $+0.43$, opposite signs from the same underlying data. The cause is Jensen’s inequality: $\mathbb{E}[100\sqrt{RV}] \neq 100\sqrt{\mathbb{E}[RV]}$ when RV is highly dispersed within the window being averaged, so the linear and quadratic averages of the same gap can disagree in sign once you narrow the window to a volatile sub-period. This doesn’t affect the full-sample statistics in Table 1. It’s just why this paper reads regime-level comparisons from both columns instead of trusting either one on its own.

4.3 The VIX Futures Term Structure

Contract definition. VIX futures are cash-settled contracts (Cboe Futures Exchange, 2026). At expiry they settle to a special VIX calculation called the SOQ, short for “special opening quotation.” Basically, this is the market’s 30-day forward-looking implied volatility measure on that settlement date. Contracts trade on a monthly cycle. Each expiry falls on the Wednesday that is 30 days prior to the third Friday of the following calendar month. This is the same convention used to set the corresponding SPX option expiry.

The key conceptual point. People often misread a longer-dated VIX future as “today’s VIX, just for a longer horizon.” That’s wrong. $F_{n,t}$, the futures price for the n -th nearest contract observed at date t , is the risk-neutral expected value of the VIX itself at that contract’s future expiry date. It is not a longer-horizon implied volatility computed from today’s option surface.

So the VIX futures curve at any date is really a strip of forward expectations of a future implied-volatility level, one expectation per expiry. It’s actually one layer more indirect than that, since the

VIX at expiry n is itself already a risk-neutral expectation of variance over the 30 days following that expiry.

This distinction matters when interpreting the curve’s shape below. A downward-sloping curve means the market prices a lower expected VIX level at the more distant settlement date than at the near one. That’s a different claim from saying implied vol is ”expected to fall over the next 30 days” as a forecast.

Construction. At each trade date, live VIX futures contracts are ranked by time to expiry: $F1_t$ is the price of the nearest-to-expiry contract, $F2_t$ the next, and so on. The dataset spans 2007–2026 rather than the full 2004–2026 futures history, since early contract-level liquidity from 2004–2006 was thin enough to introduce unreliable settlement prices; the data notes in Section 5 document a disclosure change in the CBOE settlement methodology around March 2013 and the roll convention adopted to handle contract switches near expiry.

Slope, contango, and backwardation. Define the term-structure slope as

$$S_t = F2_t - F1_t.$$

$S_t > 0$ is **contango** (the curve slopes upward: more distant expiries price a higher expected VIX level); $S_t < 0$ is **backwardation** (the curve slopes downward, typically observed during acute market stress, when the near-term VIX spikes above longer-dated expectations of it).

The economics. In a contango regime, a short position in the near-dated future tends to earn a positive roll-down return. As expiry approaches, the contract’s price slides down the curve toward the spot VIX level, which is on average lower, assuming the curve’s shape doesn’t itself shift against you in the meantime.

This roll-down return is really just the futures-market version of the same premium measured directly in Section 4.1. Both are compensation for being willing to hold short volatility exposure, whether through a variance swap payoff or through the shape of the futures curve.

The two should move together. Section 5 backs that up directly. Contango holds 82% of the time over the 2007–2026 futures sample, which closely tracks the 84% frequency of positive daily VRP over the longer 2004–2026 sample. These are two measurements of the same underlying premium, taken through different instruments and over slightly different windows.

4.4 Who Earns It, and Why It Persists

The insurance framing from 4.1 can be stated more precisely. Being short variance, whether through a short variance swap, a covered strip of short options, or a short position in VIX futures during contango, is a payoff concentrated in the high-marginal-utility states identified in 4.1. It pays steadily in calm markets and loses sharply, all at once, in crashes. The mean positive premium documented in Section 5 is compensation for bearing this asymmetric exposure. It isn’t a free lunch. The fat left tail in the short-vol backtest is the cost side of the same trade, and you can’t separate the two. If you collect the premium without ever facing that tail, you’re running a different strategy than the one measured here.

Why isn’t the premium competed away? In a frictionless market with unlimited capital, a premium this persistent and well-documented should attract arbitrage capital until it vanishes. It doesn’t, and the limits-to-arbitrage story explains why it hits short-volatility strategies especially hard. Positions get marked to market daily, so they can rack up large, visible losses during exactly the crash episodes that later prove the strategy’s premium was real, and that forces leveraged sellers to post more margin at the worst possible moment. Sharp, sudden drawdowns can trigger forced deleveraging or fund closures well before volatility has any chance to mean-revert. And traders or funds running a visibly short-vol book carry real career and reputational risk if they get caught

exposed in a crash, which scares off the very capital that would otherwise arbitrage the premium toward zero. This is the standard limits-to-arbitrage logic of Shleifer and Vishny (1997), applied here to volatility specifically; see also Ilmanen (2011) on how persistent premia in general survive because they carry rare, painful downside.

Section 5 puts each of these claims to the data. That means the premium’s magnitude and sign frequency, how it behaves across volatility regimes, the term-structure mirror relationship, and the tail outcomes that are its real economic cost.

5 Empirical Evidence

5.1 Data

Data sources: CBOE/SPX daily closing prices and CBOE VIX index level (sample 2004–2026 for the VRP construction); CBOE Futures Exchange (CFE) VIX futures settlement prices (sample 2007–2026 for the term-structure construction), incorporating the March 2013 CBOE settlement-methodology disclosure and the roll convention described in Section 4.3.

The March 2013 disclosure change introduces a short data gap of nineteen dates in the futures series. These dates are dropped from the term-structure sample instead of interpolated.

5.2 Implied vs. Forward Realized Variance

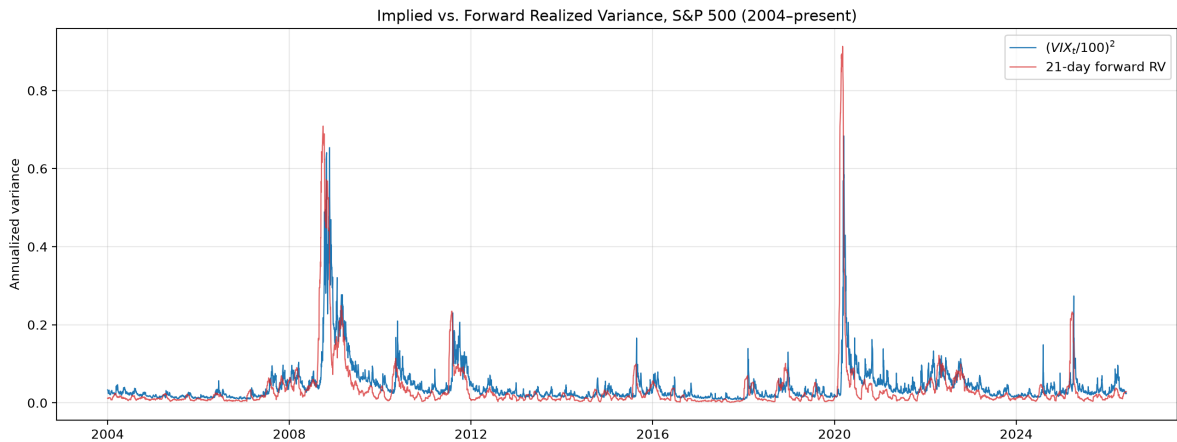


Figure 1: Implied variance ($VIX_t^2/100^2$) versus forward realized variance ($RV_{t \rightarrow t+21}$), 2004–2026, with recession/crisis periods shaded. The lead-lag pattern around major drawdowns — forward RV spiking ahead of VIX_t — serves as an internal consistency check on the forward-window construction (Section 3.4).

5.3 The Distribution of the VRP

The distribution is sharply concentrated near zero with a long left tail, consistent with a premium that is positive on 84% of sample days but occasionally hit by large negative outliers, driving the skewness of -7 in variance units.

The regime breakdown lines up with the insurance framing. The mean premium turns negative during the GFC, -0.011 in variance units, versus a positive mean in every other regime, which

Statistic	Variance units	Vol points
Mean	0.0077	3.5
Median	0.0108	4.30
Skewness	−7.43	−3.86
Positive days	84.1%	
n (observations)	5663	

Table 1: Summary statistics for daily VRP_t , 2004–2026.

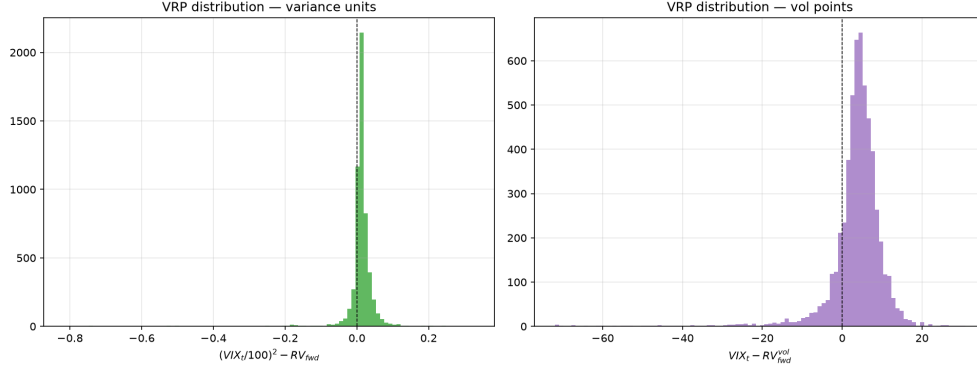


Figure 2: Distribution of daily VRP_t , 2004–2026. Note the pronounced left skew (skewness ≈ -7 in variance units) and the caveat that overlapping daily observations induce strong autocorrelation (Section 4.2); the histogram should not be read as a set of independent draws.

Regime	N days	Mean VRP (var)	Mean VRP (vol pts)	% positive
Pre-2008	986	0.008	3.14	87.3
GFC	397	−0.011	0.43	68.3
2009–2019	2644	0.011	3.91	85.0
Covid	505	0.001	4.51	85.9
Post-2022	1131	0.010	3.48	83.9

Table 2: Mean VRP_t by regime, 2004–2026.

fits with crash insurance paying out exactly when the market is under stress. The Covid regime is different. Its mean premium, 4.51 vol points, is actually higher than the calmer periods around it. Option markets priced a lot of insurance during 2020–2021 that mostly went unrealized outside the initial March 2020 shock.

5.4 The VIX Futures Term Structure

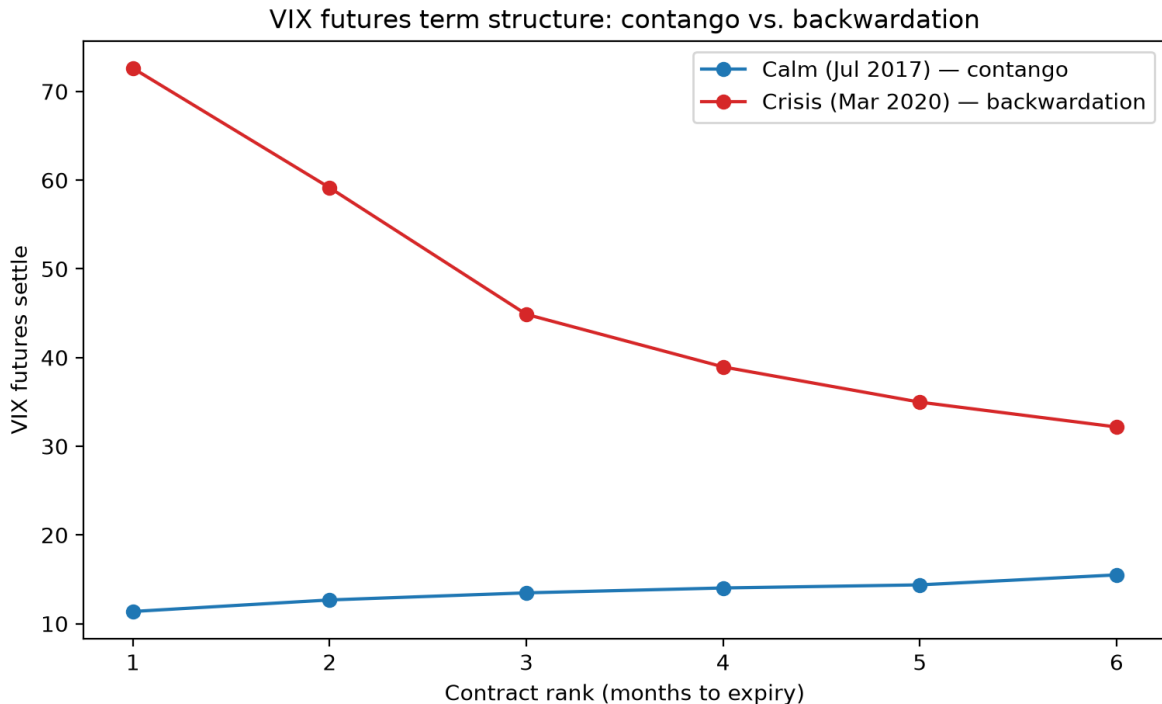


Figure 3: VIX futures term structure snapshots on selected dates, illustrating characteristic contango and backwardation shapes (Section 4.3).

The term structure is in contango on 82% of sample days (2007–2026), closely mirroring the 84% frequency of positive daily VRP reported in Table 1 (2004–2026): two instruments pricing the same underlying premium, measured over overlapping but not identical windows.

5.5 The Short-Volatility Backtest

The strategy accumulates roughly 400 vol points of gains over the sample, punctuated by two severe single-day losses of -17.6 (February 5, 2018) and -19.2 (March 16, 2020) vol points: the cost side of the premium documented above.

6 Connection to Rough Volatility

6.1 Recap of the Rough Volatility Result

The companion paper (Amargos, 2026), hereafter Paper One, shows that increments of log-volatility on European equity indices behave empirically like a fractional Brownian motion with Hurst exponent $H \approx 0.1$. That’s an order of magnitude rougher than the $H = 1/2$ built into any classical,

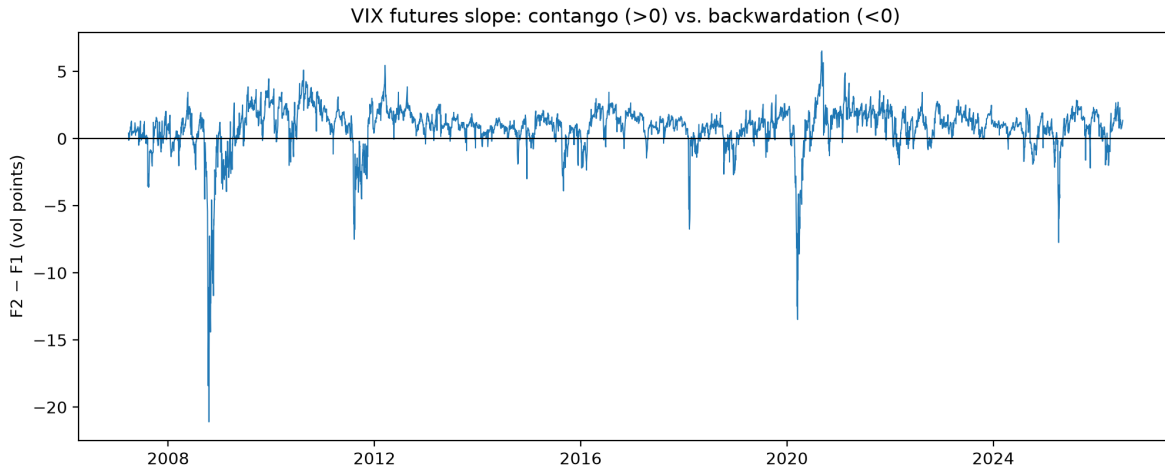


Figure 4: Term-structure slope $S_t = F2_t - F1_t$ over time, 2007–2026, with contango/backwardation regimes indicated. Contango prevails on 82% of sample days.

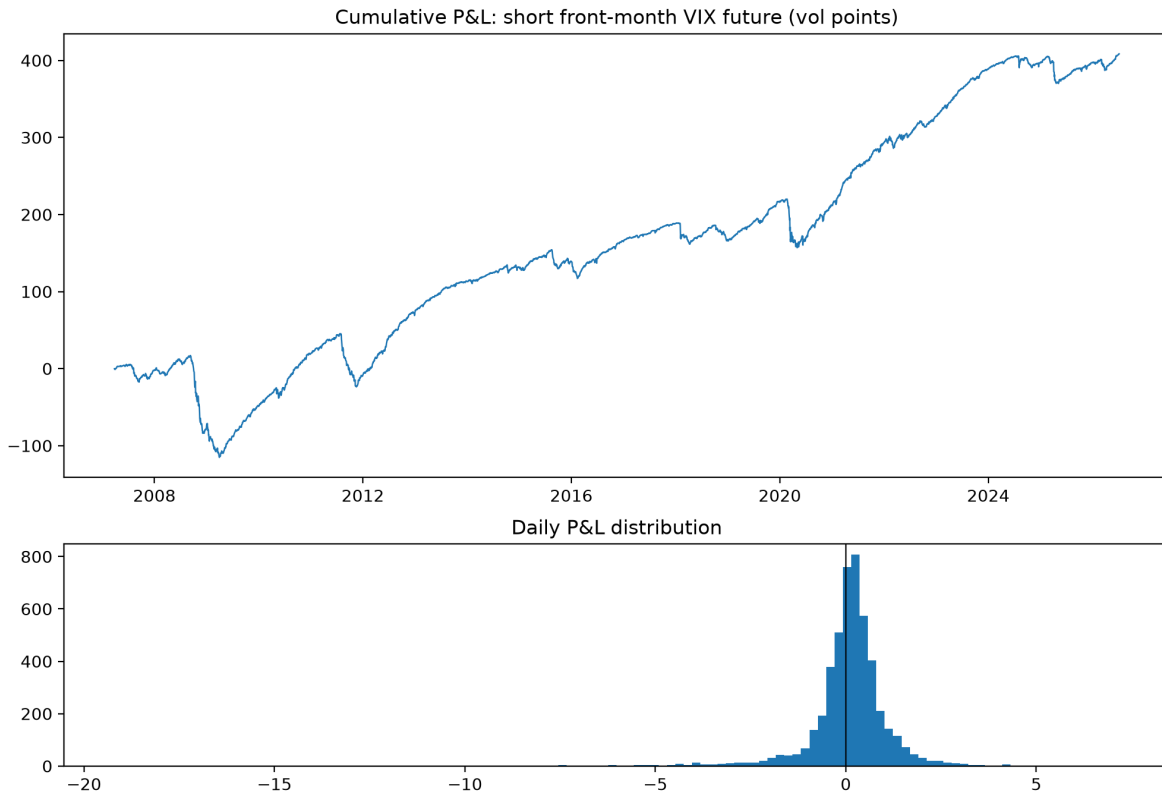


Figure 5: Cumulative P&L (vol points) of a short- $F1$, roll-at-expiry strategy, no transaction costs, 2007–2026. This shows the shape of the premium’s payoff; see Section 4.4 and Section 7 for why it isn’t a tradeable track record. Case studies: February 5, 2018 (−17.6 vol points) and March 16, 2020 (−19.2 vol points).

Markovian, Brownian-driven stochastic volatility model, and it lines up with the original finding of Gatheral et al. (2018) across a range of underlyings. Paper One’s own contribution here was a Davies-Harte simulation study validating the estimation methodology. Simulated fBm paths with known H were recovered by the estimator with bias no larger than 0.002. That gives some confidence that the empirical $H \approx 0.1$ finding on real index data reflects an actual property of volatility, rather than just an artifact of the estimator itself.

This result actually belongs here, and it’s more than just a cross-reference. Sections 2–5 measured a premium, the gap between implied and realized variance, and treated implied variance at each maturity as a single number to compare against its realized counterpart. But implied variance isn’t really a single number. It’s a curve across strikes and maturities, and the shape of that curve, especially at short maturities, is governed by the roughness of volatility. Roughness describes how σ_t moves. The VRP describes how the market prices the outcomes of that movement. That’s where the two papers connect. Roughness constrains what the short end of the implied variance term structure can look like, and that in turn shapes the F1–F2 slope of the VIX futures curve, the object whose economics Section 4 already covered.

6.2 What Roughness Implies for the Short End

The central result connecting path roughness to the shape of the implied volatility surface is that rough volatility models generate an at-the-money skew that explodes as maturity shrinks, scaling as

$$\text{ATM skew}(T) \sim T^{H-1/2}, \quad T \rightarrow 0,$$

a power-law divergence established by Bayer et al. (2016), with a rigorous short-maturity asymptotic treatment given by Fukasawa (2011, 2017). With $H \approx 0.1$, the exponent $H - 1/2 \approx -0.4$ is sharply negative. That produces a skew that steepens fast as $T \rightarrow 0$, which matches the well-documented power-law steepening seen in short-dated equity skew empirically. Classical stochastic volatility models with $H = 1/2$, Heston and its relatives, can’t reproduce this on their own. They can only get close by artificially inflating vol-of-vol parameters, which then misprices other parts of the surface. This is basically the main empirical motivation behind the rough volatility literature: roughness alone, just one parameter, fits a term-structure-of-skew fact that classical models need several parameters to approximate badly.

Implication for the objects measured in this paper. This short-end sensitivity carries over to the objects measured in this paper. It shows up whether you look at the CBOE-style variance-swap curve from Section 2 or the VIX futures term structure from Section 4. A rough volatility process reacts to a recent realized move, say a jump in recent realized variance, with a sharp adjustment right at the short end of the implied surface. That can look like fast mean reversion, but there’s no explicit mean-reverting term with a small time constant anywhere in the model. The roughness itself is what makes it look reactive over short horizons. This matters for the F1–F2 slope from Section 4.3, and specifically for how it behaves around vol spikes. F1 sits closest to the rough, reactive short end of the surface. So it should move sharply when a realized shock hits. F2 is further from where roughness dominates the surface’s local behavior, so it should move less. This gives a dynamical explanation for something Section 4 and Section 5 already show empirically: the term structure flattening or flipping into backwardation during stress episodes. Every claim in this paragraph is qualitative, and it’s grounded in the literature. There’s no new derivation here as none is needed. The point is simpler than it sounds: a known theoretical mechanism, roughness, lines up with an empirical pattern this paper already documented on its own.

6.3 What Roughness Does Not Explain

The boundary of this connection matters. Overstating it is a common mistake in this literature. Roughness, as established in Paper One, describes the **regularity and dynamics of σ_t under the physical measure $\mathbb{P}$** . In terms of the implied surface, that means the shape of derivatives prices consistent with a given rough specification. Basically, it's a statement about how volatility moves.

The variance risk premium is different. It's the wedge between two measures, $\mathbb{P}$ and $\mathbb{Q}$, applied to the same quadratic variation, as established explicitly in Sections 3.1 and 4.1. That wedge gets priced by a stochastic discount factor, or put another way, by the market's risk aversion to the specific states where variance spikes. It's a statement about how the market compensates for bearing the risk of how volatility moves, not about the movement itself. Path regularity is separate from all this. Whether σ_t is rough, smooth, or somewhere in between is, in principle, something that can be shared identically under both $\mathbb{P}$ and $\mathbb{Q}$. A well-specified model could have the same H under both measures and still generate an arbitrarily large or small risk premium, depending entirely on the pricing kernel. So roughness and the risk premium are, in this precise sense, **orthogonal**. Knowing H tells you nothing by itself about the sign or magnitude of $\mathbb{E}^{\mathbb{P}}[VRP_t]$. Measuring a large VRP empirically tells you nothing by itself about how smooth the volatility path behind it actually was.

The two papers complement each other rather than one explaining the other. Paper One described how volatility moves. This paper measured what bearing its risk actually costs. Neither result follows from the other.

An open direction, out of scope here. One natural next step, not attempted in this paper, would be a rough volatility model with an explicit specification of the market price of volatility risk. This would be a rough analogue of the volatility risk premium parameter used in classical stochastic volatility option pricing. It would need to be calibrated to match both the short-maturity skew term structure from 6.2 and the level and term structure of the variance risk premium from Sections 4–5. A model like that would unify the dynamical and economic sides of this research program into a single specification. That's a substantial modeling project on its own, and it's left for future work here.

6.4 Bridge to the Conclusion

Taken together, the two papers form a research program in volatility, split along the line drawn in 6.3. Paper One covers the dynamics: how volatility actually moves, rougher empirically than any classical model assumes. This paper covers the economics: what the market charges to shift the risk of that movement from one party to another. Together they give a fuller picture of volatility than either paper does alone. The connection made in this section, roughness shaping the short end of the term structure whose economics this paper measured, is where the two meet. Beyond that, each paper stays in its own lane.

7 Conclusion

Option-implied variance on the S&P 500 exceeds realized variance on most trading days. The average gap is about 3.5 volatility points, positive on 84% of trading days across the 2004–2026 sample. The distribution of this gap is sharply left-skewed, and its sign inverts in crisis: it turns negative during the GFC, the one regime in the sample where crash insurance actually got called on. The VIX futures term structure tells the same story through a different instrument. Contango

holds on 82% of days, which closely mirrors the 84% positive-VRP frequency measured directly. Put together, these facts describe a market that pays a steady premium for insurance against variance, and that premium goes to whoever is willing to sell it.

Getting to that measurement meant establishing, step by step, that variance itself is a replicable object and not just a statistic. Itô’s lemma applied to the log price shows that realized variance is the payoff of a specific dynamic-hedging strategy against a static short log contract. The Carr-Madan spanning identity shows that the static leg of that strategy, the log contract, can itself be replicated by a continuum of listed options. The CBOE’s VIX is, once you account for four named and reconciled discretization adjustments, the discretized market version of that same replicated object. And the forward-window convention, comparing VIX_t against realized variance over $[t, t+21]$ rather than the trailing window, is what makes the resulting comparison actually say something about a risk premium instead of just reflecting volatility clustering. Every link in this chain had to hold for the headline numbers above to mean what they’re claimed to mean.

Several limitations affect how much weight these results can carry. The theoretical derivation assumes a continuous price process with no jumps. The Carr-Wu jump correction would be the right extension, and this paper leaves it unimplemented. Realized variance is estimated from daily close-to-close returns, which is noisier than intraday alternatives. It was chosen for robustness and data availability over a 22-year sample instead of statistical efficiency. The short-volatility backtest carries no margin requirements, transaction costs, or position sizing rules. It illustrates the shape and sign of the premium’s economic payoff. It is not a tradeable strategy net of real-world friction. Daily VRP observations overlap in their underlying returns and are therefore strongly autocorrelated, which limits formal statistical inference on the series to methods built to handle that overlap. And the sample itself has a data gap of nineteen dates around the March 2013 CBOE settlement-methodology disclosure. That gap is documented in the data appendix, not filled in.

For a practitioner, it comes down to this: selling volatility is selling insurance. The premium documented here is real, and it held up across two decades and multiple market regimes, paid for a specific reason: compensation for bearing a risk concentrated in exactly the states investors care most about avoiding. Anyone collecting this premium is exposed to the left tail and earning the roll-down described in Section 4. The two single-day losses documented in Section 5, February 5, 2018, and March 16, 2020, aren’t outliers to wave away. They show what that tail actually costs when it shows up.

One clear next step for this research program is sketched here, though not attempted. The most direct extension would be a rough volatility model with an explicit market price of volatility risk. Such a model could jointly fit the short-maturity skew term structure documented in the companion paper and the level and term structure of the variance risk premium measured here. That would unify the dynamical and economic sides of this work in a single specification. Beyond that, the same measurement apparatus extends naturally to variance risk premia in interest rate and currency markets. And the realized-variance side of the construction could be sharpened with intraday, noise-corrected estimators, wherever high-quality data allows it.

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